

Machine Learning for Dynamic Bitcoin Allocation: A Risk-Adjusted, Long-Only Strategy

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Confidentiality note. This is a public document. Confidential partner information - assets under management, client data, and proprietary implementation details (exact feature set and hyperparameters) - has been intentionally omitted. Performance figures reported here are **out-of-sample backtest and early paper-trading results**, not realized returns on capital.

Abstract

We present a machine-learning system that decides, on a weekly basis, how much of a portfolio to allocate to Bitcoin (BTC) versus a cash-equivalent benchmark (the Brazilian interbank rate, CDI). The system is **long-only** (allocations between 0% and 100%, no short selling) and is explicitly optimized for **downside risk-adjusted return** rather than raw return. Its core is a dual ensemble of gradient-boosted decision trees that forecasts short-horizon Bitcoin returns; the forecast is translated into a position size that is scaled by a market-regime filter and by the model's own confidence, and then bounded by automated risk controls. In a 4.3-year out-of-sample backtest, the strategy achieved substantially higher downside-adjusted returns than static allocations while cutting the maximum drawdown to roughly one-tenth of a buy-and-hold position. A battery of overfitting tests - including walk-forward validation with temporal purging, deflated performance ratios, and label-permutation tests - supports the conclusion that the signal is statistically real. The system is currently in a paper-trading validation phase.

1. Introduction and problem

Bitcoin offers a long-run risk premium but with extreme volatility: peak-to-trough losses of 65-80% are common. For institutional vehicles and most investors, that drawdown profile is hard to hold. In practice, "how much Bitcoin to carry" is often a **discretionary** decision - difficult to audit, to reproduce, and to scale.

This project addresses a focused question: **can the Bitcoin risk premium be captured with a fraction of its drawdown, through a systematic, auditable process rather than discretionary judgment?** The objective is not to maximize return at any cost, but to maximize *risk-adjusted* performance under a long-only mandate.

2. Objectives

1. Build a systematic weekly BTC/CDI allocation signal that **dominates static allocations** on a downside-risk-adjusted basis.
2. Keep the maximum drawdown materially below a buy-and-hold position.
3. Ensure the process is **reproducible, validated against overfitting, and auditable** - suitable for review by a risk committee.

4. Operationalize the signal as a one-command production pipeline, and validate it in paper trading before any capital is committed.

Success is measured with portfolio-level, risk-adjusted metrics (Sortino ratio, maximum drawdown, excess return versus benchmarks), not with classifier accuracy in isolation - by design, the model's edge comes from *sizing* positions well, not from predicting direction frequently.

3. Methodology

3.1 Data

The pipeline ingests data from a dozen public and licensed sources spanning three families: **macro** (e.g., monetary aggregates and policy-rate proxies), **on-chain** (network and valuation metrics), and **market/technical** (price, volume, derivatives basis and funding). From these, the system derives roughly thirty engineered features, including trend- and mean-reversion indicators and structural-regime measures. Feature construction is strictly **backward-looking** - a property enforced by an automated test that perturbs the most recent observation and verifies that no past feature value changes, guarding against look-ahead bias.

3.2 Forecasting model

The forecasting core is a **dual ensemble** of gradient-boosted decision trees: one ensemble regresses the short-horizon (multi-day) forward return of Bitcoin, while a second classifies the direction of that move. Bagging many trees per ensemble reduces seed-to-seed variance and makes the signal more stable in live use. The model family was selected after comparing it against alternatives - including hidden Markov regime models, recurrent neural networks, random forests, and stacked meta-learners - all of which underperformed the chosen approach on the target objective.

3.3 Position sizing

The regression forecast is converted into a target allocation that is **scaled by two factors**: (i) a *market-regime filter*, derived from long- and short-term moving averages, which reduces exposure in bearish regimes; and (ii) a *confidence weight* derived from the directional classifier, so the strategy bets larger only when the model is more certain. The result is clipped to the [0%, 100%] long-only range. The portfolio rebalances weekly, with an additional emergency rebalance triggered by very large single-day moves. The model is retrained on a fixed semi-annual schedule using an expanding window.

3.4 Risk controls

Three automated controls are applied to every signal: a **kill switch** that caps exposure when cumulative drawdown breaches a threshold; an **accuracy de-risk** that halves exposure when rolling predictive accuracy deteriorates while the model remains overconfident; and a **population-stability monitor** that flags distribution drift in the input features. These controls are part of the production signal, not an afterthought.

3.5 Validation methodology

Validation follows established financial-ML practice:

- **Walk-forward, out-of-sample evaluation** with an expanding window and a temporal **purge/embargo** between train and test to prevent leakage.
- **Multi-seed** runs to quantify the variance of every reported metric.
- **Label-permutation testing**: the model is re-evaluated on randomly shuffled targets to establish a null distribution.
- **Deflated performance ratios** to discount the effect of having tried many configurations (multiple-testing).

4. Results

All figures below are **out-of-sample backtest** results over roughly 4.3 years (248 weekly rebalances), reported on a multi-seed basis and net-aware of realistic transaction costs (a few basis points per rebalance; results remain strong even under pessimistic cost assumptions).

Strategy	Sortino (daily)	Max drawdown	Outcome
Cash benchmark (CDI)	-	0%	Low return, no risk
Static 30% BTC / 70% CDI	~1.1	~-22%	Baseline blend
100% Bitcoin (buy-and-hold)	~0.6	~-67%	High risk
This strategy (long-only ML)	~3.5	~-7%	Best risk-adjusted

The strategy delivered markedly higher downside-risk-adjusted performance than either static allocation, while holding maximum drawdown to roughly **one-tenth** of a buy-and-hold position - a direct result of spending a majority of the time in the cash benchmark and concentrating exposure in favorable regimes.

Statistical significance of the edge. In label-permutation testing, **none of 100 shuffled-target runs beat the baseline** ($p < 0.01$), indicating the signal is not an artifact of chance. After applying a deflated performance ratio to account for multiple-testing, the strategy still passes under a realistic estimate of the number of effective trials.

Early live (paper) evidence. In a strict out-of-sample window of the most recent year-to-date (about 105 days), the strategy returned **+19.7%** while Bitcoin buy-and-hold returned **-14.4%** and the cash benchmark **+4.1%** - early corroboration, though concentrated in a small number of rebalances.

5. Discussion and limitations

We are deliberately conservative about these results:

- **Deflation.** Backtest metrics overstate live expectations. After discounting for multiple-testing, the *realistic* forward expectation is a Sortino in the 1.5-2.5 range and a drawdown of -15% to -25% - still superior to the static baselines, but well below the raw backtest.
- **Concentration.** A small number of weeks account for a large share of the cumulative return; missing a few of them would materially reduce the edge.

- **Regime risk.** The model depends on periodic retraining; an abrupt structural shift in market behavior could outpace it, which is partly why the automated risk controls exist.
- **Status.** The system is in **paper trading**; reported numbers are not realized returns on capital. A decision gate evaluates the strategy against conservative, pre-registered thresholds before any capital is committed.

Ethics and sustainability. The model uses only market and on-chain data - no personal data - so fairness/bias concerns are minimal. Its computational footprint is very low (CPU-only, retrained twice a year). As a credibility safeguard, the project adopts an explicit honesty policy: results are always labeled as backtest/paper-trade and reported with their deflated, conservative range.

6. Conclusion and next steps

This project demonstrates that a disciplined machine-learning approach can turn a discretionary "how much Bitcoin to hold" decision into a **systematic, auditable, and risk-controlled** process that, in out-of-sample testing, substantially improves downside risk-adjusted returns and sharply reduces drawdown relative to static allocations. The evidence that the signal is statistically real - surviving permutation and deflation tests - distinguishes it from an overfit backtest.

Next steps: complete the paper-trading window and pass the decision gate before committing capital; add model-explainability tooling; and, conditional on the gate, invest in production hardening and extend the same framework to additional asset/cash pairs.

References

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